



Geospatial Terminals Operations (GTO) Automated Charting and Encoding (ACE) Version Description Document

Foundation Sustainment Services (FS2)

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Revision History

Version #	Implemented By	Revision Date	Reason
1.0.28	MDA	12/09/24	Initial release of this version of the product.

Authorization Notification

Any authorization applies only to the current version of the document.s

Role	Name	Company	Job Title
Prepared By			Mission Partner
Reviewed By			Mission Partner
Checked By			CM Lead
Authorized By			Program Manager

Approval

The undersigned have read this VDD and approve its contents.

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Date



Contents

1	Scope	4
1.1	Introduction	4
1.2	System Overview	4
1.3	Document Overview	5
2	Version Description.....	6
2.1	Deployment Environment	6
2.2	Software Functionality and Capabilities	8
2.3	Inventory of Software Components.....	8
2.3.1	Software Deliverables.....	8
2.3.2	Software List.....	8
2.4	Changes Installed	10
2.5	Installation Instructions	10
2.6	Possible Problems and Known Errors	11



1 Scope

1.1 Introduction

This document is the *Version Description Document (VDD)* for *Geospatial Terminals Operation (GTO) Program-Aeronautical Automated Charting and Encoding (ACE) Software version 1.0.28*.

This VDD details the software components to be delivered with ACE Version 1.0.28. This document describes the features included in the delivery of the software build, at entrance to test. It includes the build artifacts for deployment, any known issues with the delivery, and the changes included in the software version.

1.2 System Overview

Automated Charting and Encoding (ACE), is a software product that the NGA has licensed from MDA for the GTO program. ACE is a next-generation aeronautical chart management solution that provides fully automated chart production directly from aeronautical data sources. Publication-quality charts are generated without human intervention, and in a wide range of user-defined templates and styles. An intelligent deconfliction capability virtually eliminates the need for user refinements to a chart. Any refinement (e.g. repositioning a label) is retained by ACE and automatically applied to future chart versions. Using an intuitive and interactive web-based user interface, ACE provides highly automated processes that require minimal training, while substantially reducing maintenance costs.

In the following diagram, the input data set includes the ADDE database, AFD data, Vertical Obstructions (VO'S), Chart data and extents, Digital Terrain & Elevation Data (DTED), and Water Feature Tiles.

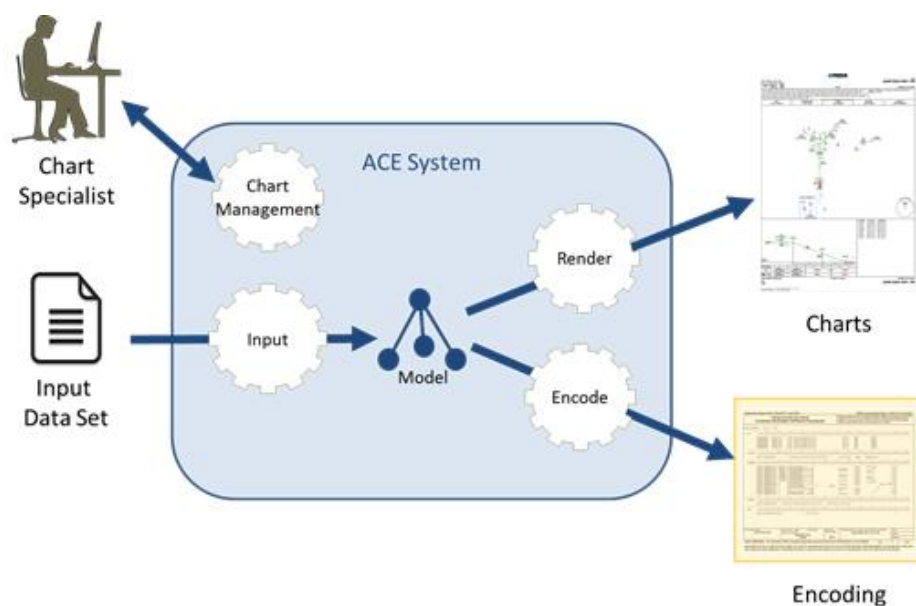


Figure 1: An overview of the ACE System



1.3 Document Overview

The ACE VDD provides the list of software delivered for Version 1.0.28 (2412) of the product and the capabilities of that software. This document details a single release of the product and will be updated only to correct information specific to version 1.0.28 of the product. Follow-on versions of the product will be documented in a separate VDD specific to that version of the product.

The document is organized as follows:

- Section 1 – Scope
- Section 2 – Version Description



2 Version Description

This release contains the following deliverable packages for all ACE software components:

- NGA Production Source Code (ZIP file)
- GTO Production Chart Management (ZIP file)
- Microfocus Fortify (scan files)
- GTO ACE Web Deployment Installation Notes (PDF file)
- GTO ACE Release Notes (PDF file)
- SSD Specification (HTML file)

2.1 Deployment Environment

ACE is installed on a server in NGA's COE environment. The following pre-requisite software packages must be installed on GTO servers before installing ACE.

- Windows 2016 R2 Server or higher
- PostgreSQL – Version 14.4.X or higher
- PostGIS – Version 3.X or higher
- Adobe Acrobat Reader – Version 2023.X or higher
- Google Chrome – Version 117.X or higher
- Apache Web Server – Version 2.4.57 or higher
- Java Runtime Environment 8 – 1.8.381 or higher



Architecture Diagrams

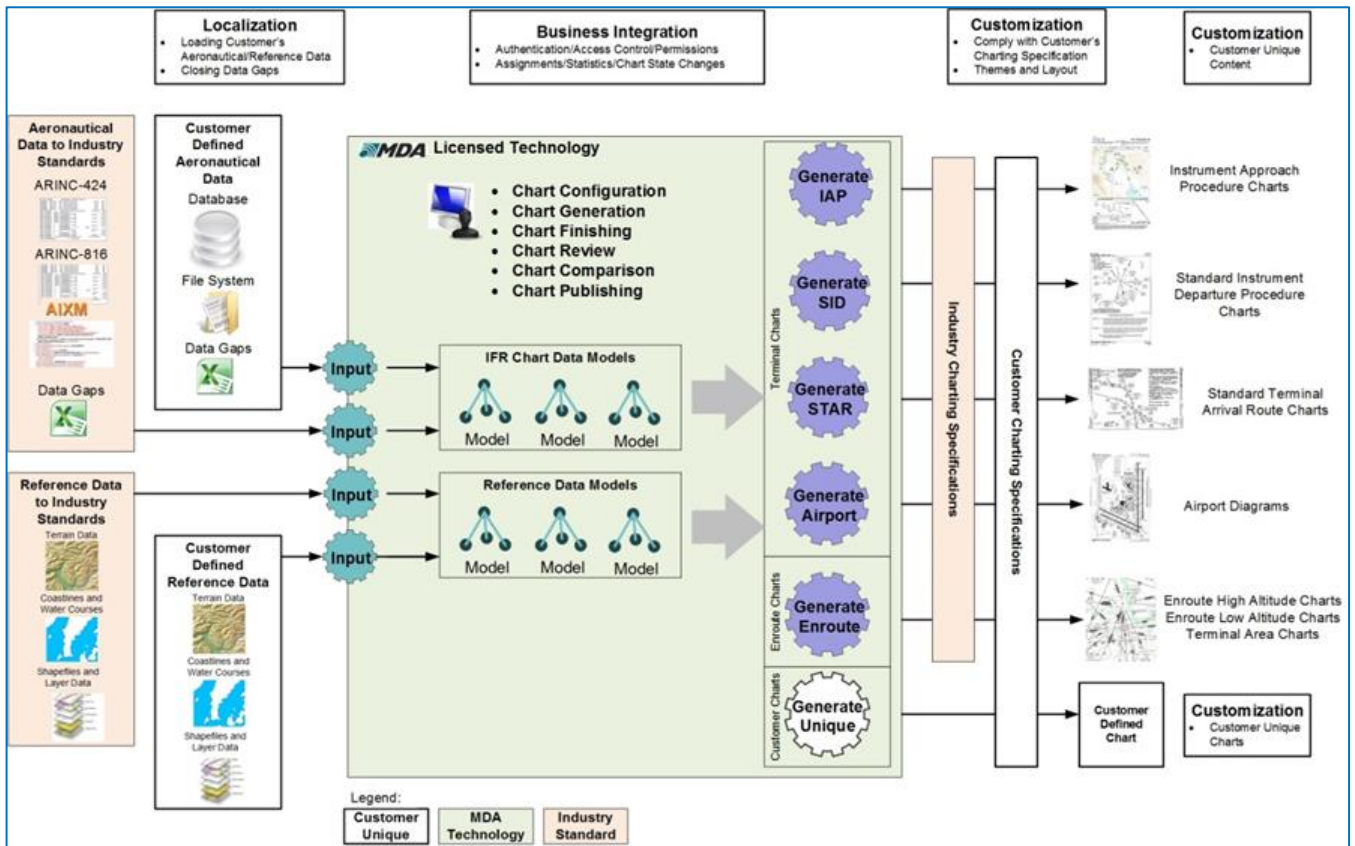


Figure 2: A sample ACE architecture.

Aero GTO As-Is Architecture Diagram Version 1.3

Date: 1/25/2024

NOTE: All listed machines, databases, and virtual machines are hosted on COE

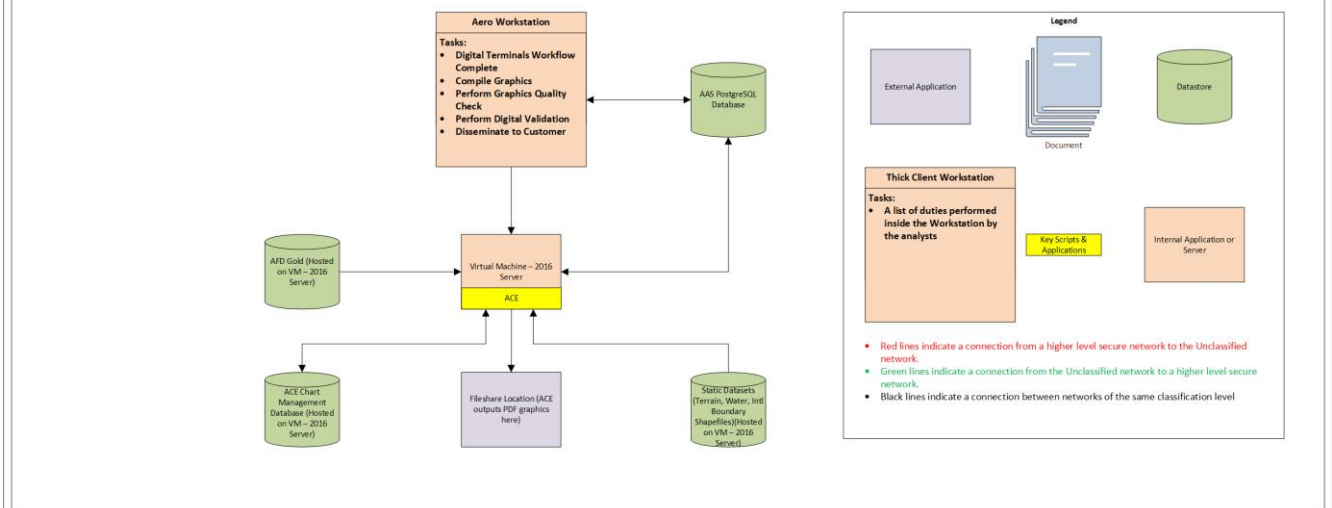


Figure 3: SFA As-Is Architecture Diagram.



2.2 Software Functionality and Capabilities

The following list identifies new or modified capabilities delivered by this version of ACE software.

ACE Version 1.0.28 (2412)

Domain	Feature ID	Feature Name	Summary of Feature
	AUTOCHART-19718	Html::Tiny	Replace CGI module with Html::Tiny
	AUTOCHART-19718	Html::Escape	Replace CGI module with Html::Tiny

2.3 Inventory of Software Components

This section documents the software configuration for ACE.

2.3.1 Software Deliverables

MDA delivers software packages and build artifacts containing the application components, including but not limited to Perl files and JavaScript, HTML, and CSS scripts. These packages will include code library dependencies along with custom code to ensure that the application has all of the software needed to execute.

2.3.2 Software List

The ACE 1.0.28 (2412) Software List can be found in Table 2.3.2.

Table 2.3.2 - Software List

Software	Version	SWID	Vendor	Control #	Notes
Automated Charting and Encoding (ACE)	1.0.28	6551	MDA (MacDonald, Dettwiler and Associates Ltd)	20241219054	
Perl	5.32.1	4781	Strawberry Perl	20210311017	
Batik and Dependencies					
Batik	1.17	7265	Apache Software Foundation	20240131067	
UI NPM Packages					
@angular/animations	15.2.10	15809	Google	20240326025	
@angular/common	15.2.10	16663	Google	20231120067	
@angular/compiler	15.2.10	16668	Google	20231120068	
@angular/core	15.2.10	30841	Google	20241113048	
@angular/forms	15.2.10	16764	Google	20231120071	
@angular/platform-browser	15.2.10	16675	Google	20231120073	
@angular/platform-browser-dynamic	15.2.10	16676	Google	20231120074	
@angular/router	15.2.10	16677	Google	20231120075	
@angular/upgrade	15.2.10	19791	Google	20241113045	
file-saver	2.05	23566	Eli Gray	20210113034	
ngx-bootstrap	10.2.0	15789	Valor Software	20230524011	
rxjs	6.6.7	12649	Ben Lesh	20210718009	
tslib	2.5.0	19228	Microsoft	20230504055	
zone.js	0.11.8	13773	Google	20221004097	



3 rd Party Perl Packages					
<u>Direct Uses</u>			All distributions listed below maintained at https://metacpan.org/		
Compress::Raw::Lzma	2.103	34368	Paul Marquess (PMQS)	202502210023H	
Config::IniFiles	3.000003	34369	Shlomi Fish (SHLOMIF)	202502210025H	
Crypt::PBKDF2	0.161520	34370	Andrew Rodland (ARODLAND)	202502210026H	
Crypt::Rijndael	1.16	34371	Leon Timmermans (LEONT)	202502210027H	
Data::UUID	1.226	34372	Ricardo SIGNES (RJBS)	202502210028H	
Date::Calc	6.4	34373	Steffen Beyer (STBEY)	202502210029H	
DateTime	1.58	34375	Dave Rolsky (DROLSKY)	202502210031H	
DateTime::Format::Strptime	1.79	34376	Dave Rolsky (DROLSKY)	202502210033H	
DBIx::Log4perl	0.26	34377	Martin J Evans (MJEVANS)	202502210034H	
File::Next	1.18	34378	Andy Lester (PETDANCE)	202502210035H	
Filesys::DfPortable	0.85	34380	Ian Guthrie (IGUTHRIE)	202502210037H	
Filter::Crypto	2.09	34381	Steve Hay (SHAY)	202502210038H	
Font::TTF	1.06	34382	Bob Hallissy (BHALLISSY)	202502210039H	
Font-TTFMetrics	0.1	34383	Malay Kumar Basu (MALAY)	202502210041H	
Geo::Ellipsoid	1.15	34384	Peter John Acklam (PJACKLAM)	202502210043H	
Geo::ShapeFile	3.01	34386	Shawn Laffan (SLAFFAN)	202502210045H	
HTTP::Message	6.36	34388	Olaf Alders (OALDERS)	202502210047H	
HTML::Escape	1.11	34389	MATSUNO★Tokuhiro"<xmp>(TOKUHIROM)	202502210048H	
HTML::Tiny	1.08	34390	Andy Armstrong (ARISTOTLE)	202502210049H	
JavaScript::Minifier	1.16	34391	Zoffix Znet (ZOFFIX)	202502210050H	
JSON	4.06	34392	Kenichi Ishigaki (ISHIGAKI)	202502210051H	
JSON::WebToken	0.10	34394	Yuji Shimada (XAICRON)	202502210053H	
libwww-perl	6.66	34395	Olaf Alders (OALDERS)	202502210054H	
Log-Dispatch-File-Rolling	1.09	34396	Michael Jacob (JACOB)	202502210055H	
Log::Log4perl	1.54	34397	Ed J (ETJ)	202502210056H	
Math::Clipper	1.29	34398	Michael E. Sheldrake (SHELDRAKE)	202502210057H	
Math::Polygon	1.10	34399	Mark Overmeer (MARKOV)	202502210058H	
Mojolicious	9.26	34401	Sebastian Riedel (SRI)	202502210060H	
Moose	2.2201	34402	Karen Etheridge (ETHER)	202502210060H	
PDF::API2	2.038	34403	Steve Simms (SSIMMS)	202502210062H	



Readonly	2.01	34405	Sanko Robinson (SANKO)	202502210063H	
Scalar::List::Utils	1.62	34406	Paul Evans (PEVANS)	202502250130H	
String::Util	1.32	34407	Scott Baker (BAKERSCOT)	202502210066H	
Tree::R	0.072	34409	Ari Jolma (AJOLMA)	202502210068H	
Try::Tiny	0.31	34410	Karen Etheridge (ETHER)	202502210069H	
URI::Escape	5.10	34411	Olaf Alders (OALDERS)	202502210070H	
Win32::Shortcut	0.08	34414	Jan Dubois (JDB)	202502210073H	
XML::LibXML	2.0207	18556	Shlomi Fish (SHLOMIF)	202502210074H	
XML::XPath	1.47	34415	Mohammad S Anwar (MANWAR)	202502210075H	
<i>Dynamic Dependencies</i>					
Parse::Binary	0.11	34416	Audrey Tang (AUDREYT)	202502210076H	
Win32::Exe	0.17	34417	Mark Dootson (MDOOTSON)	202502210077H	
Filter::Crypto::Decrypt	2.09	34418	Steve Hay (SHAY)	202502210078H	
PAR	1.017001	34419	Roderich Schupp (RSCHUPP)	202502210079H	
autodie::exception	2.34	34420	Todd Rinaldo (TODDR)	202502210080H	

2.4 Changes Installed

As was noted in Section 1.3, this is considered to be in initial release of the 1.0.28 product, so changes from the previous version are not recorded in this VDD.

Table 2.4 includes defects recorded in testing as well as any other internally discovered defects that were resolved.

- Issue Types:
- Test Problem Report (TPR): DRs written by the NTO during testing
- System Problem Report (SPR): DRs written by problems found by the PMO, customer, or external systems.

Table 2.4 - Changes Installed

PBI/INC #	Key	Issue Type	Summary

2.5 Installation Instructions

ACE GTO consists of a Windows Server installation with the ACE software installed with an Apache Web Server within NGA's COE environment. The ACE application connects to external AFD and APS databases along with its' own internal ACE database.



(Additional details for manually installing and configuring the application can be found within *GTO ACE v1.0.28 Web Deployment.pdf*)

2.6 Possible Problems and Known Errors

Table 2.6 Version 1.0.28 (2412) Possible Problems and Known Errors

Issue ID	Summary	Description